7 **(a)** **(i)** Consider each mass separately,

$T - W_P = m_P a$ ---(1)

$W_Q - T = m_Q a$ ---(2)

Solving the simultaneous equations (1) + (2),

$(1.00)(9.81) - (0.50)(9.81) = (1.00 + 0.50)a$

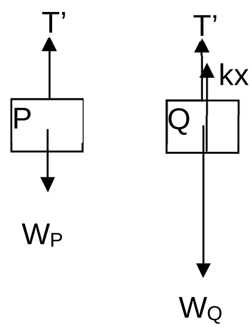
$a = 3.27 \text{ m s}^{-2}$

From equation (1) in 1(a)(i),

$T = 0.50(9.81) + 0.50(3.27)$

$= 6.54 \text{ N}$ (accept 3 or 4 s.f.)

(ii) After Block Q comes to a stop, the forces acting on P and Q are in equilibrium.



$T' = W_P$ ------------------ (3)

$T' + kx = W_Q$ ------------------ (4)

Substitute (3) into (4)

$0.50(9.81) + 200x = 1.00(9.81)$

$x = 0.0245 \text{ m}$ (3 s.f.)

(b) **(i)** By conservation of momentum,

Initial momentum of bullet = final momentum of bullet and block

$m_{bullet} \times 100 = (m_{bullet} + m_{block})v$

$v = 0.990 \text{ m s}^{-1}$

By conservation of energy,

Loss in k.e. of bullet and block = Gain in g.p.e of bullet and block

$\frac{1}{2}(m_{bullet} + m_{block})v^2 = (m_{bullet} + m_{block})gh$

$h = \frac{1}{2}(0.990)^2/(9.81) = 0.050 \text{ m}$

(ii) The collision is perfectly inelastic.

(c) **(i)** $a = \frac{v - u}{t} = \frac{24.72 - 0}{4.5} = 5.493 \text{ m s}^{-2}$

$P_{ave} = F(\frac{1}{2}v_{max}) = ma(\frac{1}{2}v_{max}) = (6500)(5.493)(\frac{1}{2})(24.72) = 441 \times 10^3 \text{ W}$

(ii) Loss in KE = Gain in GPE

$\frac{1}{2}mv^2 = mgh$

$h = \frac{v^2}{2g} = \frac{24.72^2}{2(9.81)} = 31.2 \text{ m}$

Hence it will not go beyond the track which is 45.0 m tall.

(iii) Let the speed of the train at the top of the loop be v_{top},

$$\tfrac{1}{2}mv_{top}^2 = \tfrac{1}{2}mv^2 - mgh$$
$$v_{top} = 12.6 \text{ m s}^{-1}$$

To stay on the seat (and not fall off) at the top of the inner loop, N > 0.

$$N + mg = \frac{mv_{min}^2}{r},$$
$$N = \frac{mv_{min}^2}{r} - mg$$

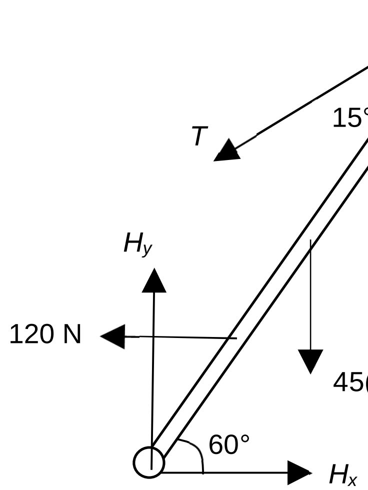
Hence, to not fall off, N > 0 and therefore $v_{min} > \sqrt{(rg)}$.

$$\sqrt{(rg)} = \sqrt{(11.5)(9.81)} = 10.6 \text{ m s}^{-1}$$

Since $v_{top} > 10.6$ m s^{-1}, the rider will not fall off if the lap-bar mechanism fail.

(d) **(i)** Net force is zero in any direction.
Net moment is zero about any point.

(ii)



Evidence of angle between *T* and boom (in working) = 15°
Taking moments about the hinge,

$(T\sin 15°)(3) + (120\sin 60°)(0.75) = (45)(9.81)(1.5\cos 60°) + (1962)(3\cos 60°)$

$T = 4116 = 4.1$ kN

(iii) (Let H_y and H_x be the vertical and horizontal components of the force hinge acts on the boom)

$H_y = 1962 + 45(9.81) + T\cos 45° = 5314$ N
$H_x = 120 + T\sin 45° = 3030$ N

Therefore, magnitude of force hinge acts on boom = $\sqrt{5314^2 + 3030^2} = 6120$ N
Direction of the force = $\tan^{-1}(5314 / 3030) = 60.3°$ above horizontal

8 (a) (i)